

AI Resume Ranking & Job Matching System

Professional Software Project Documentation

A complete technical and functional specification for an intelligent platform that evaluates resumes, recommends suitable roles, analyzes job descriptions, identifies skill gaps and provides role-specific resume optimization.

Document Information	Details
Project Type	AI-powered Web Application
Domain	Recruitment Technology / Career Intelligence
Primary Users	Students, job seekers, professionals, career coaches and placement teams
Core Functions	Resume analysis, ATS scoring, role matching, JD analysis and AI recommendations
Version	1.0
Status	MVP / Prototype with production-ready architecture

1. Executive Summary

The AI Resume Ranking & Job Matching System is a web-based career intelligence platform designed to help candidates understand the quality of their resume and determine how well their profile fits specific job opportunities. The platform combines resume parsing, structured profile extraction, scoring, role recommendation, job-description analysis, skill-gap detection and AI-assisted improvement.

The system converts an unstructured resume into measurable insights. Users can upload a resume, receive an overall score and ATS-oriented score, explore recommended roles, paste a target job description, see the match percentage and receive specific guidance on what should be improved.

The intended production product extends beyond a resume checker. It is designed as a career optimization platform that can eventually support job discovery, application tracking, interview preparation, learning recommendations and continuous profile improvement.

2. Vision and Mission

Vision

Build an intelligent career assistant that helps every candidate understand their current employability, target realistic opportunities and continuously improve their professional profile.

Mission

Use AI responsibly to transform resumes and job descriptions into clear, measurable and actionable career insights while keeping the candidate in control of the final content.

3. Objectives

- Provide a professional and simple resume-analysis experience.
- Accept PDF, DOC and DOCX resumes with validation.
- Extract skills, education, experience, projects and certifications.
- Generate resume quality and ATS-oriented scores.
- Recommend suitable job roles with match percentages.
- Analyze target job descriptions and extract requirements.
- Compare resume evidence against job requirements.
- Identify missing skills and important keywords.
- Generate role-specific improvement suggestions.
- Show application readiness and prioritized skill gaps.
- Create an architecture that can scale from MVP to production.

4. Problem Statement

Candidates often use the same resume for different jobs and do not know whether their skills, experience and wording match the requirements of a specific role. Manual comparison between a resume and a job description is slow and subjective. Important keywords may be missed, achievements may be described too generally and candidates may apply to roles that do not align with their current profile.

The proposed platform solves this problem through automated analysis and explainable recommendations, giving users a clear path from resume upload to targeted application.

5. Scope

5.1 Included

- Resume upload and validation.
- Resume text extraction and parsing.
- Resume scoring and ATS-oriented evaluation.
- Skill and profile extraction.
- Recommended role ranking.
- Job-description analysis.
- Resume-to-job matching.
- Missing skill and keyword detection.
- Role-specific resume improvement.
- Skill-gap prioritization.
- Application-readiness guidance.

5.2 Excluded from Initial MVP

- Guaranteed interview or employment outcomes.
- Automatic submission of applications to external portals.
- Fabrication of experience or qualifications.
- Automated final hiring decisions.
- Access to private job-board accounts without explicit integrations and consent.

6. Target Users

User	Main Need	Value
Student / Fresher	Understand strengths and gaps	Role suggestions and learning priorities
Job Seeker	Tailor resume to vacancies	JD matching and recommendations
Professional	Change role or industry	Transferable-skill analysis
Career Coach	Review candidate profiles	Structured evaluation
Placement Team	Evaluate many candidates	Consistent scoring workflow

7. Functional Requirements

ID	Requirement	Description	Priority
FR-01	Resume Upload	Accept supported resume files and validate them.	High
FR-02	Resume Parsing	Extract structured candidate information.	High
FR-03	Resume Scoring	Calculate overall and component scores.	High
FR-04	Role Recommendation	Rank suitable job roles.	High
FR-05	JD Analyzer	Extract skills, keywords and requirements.	High
FR-06	Job Matching	Compare candidate profile with target job.	High
FR-07	Gap Analysis	Identify missing or weak requirements.	High

ID	Requirement	Description	Priority
FR-08	AI Suggestions	Generate targeted resume improvements.	High
FR-09	Dashboard	Present insights through a visual UI.	High
FR-10	Analysis History	Store previous analyses for comparison.	Medium

8. Non-Functional Requirements

- Usability: clear navigation and understandable results.
- Performance: responsive UI and visible analysis progress.
- Security: secure transport, authorization and protected documents.
- Scalability: independently scalable analysis services.
- Reliability: graceful recovery from invalid files and service failures.
- Maintainability: modular frontend, backend and AI layers.
- Accessibility: readable typography, sufficient contrast and accessible controls.
- Privacy: clear retention, deletion and processing policies.

9. System Architecture

The recommended production architecture separates presentation, application logic, document processing, AI orchestration and persistence. This separation allows the initial interface to evolve without exposing secrets or coupling the browser directly to AI services.

Layer	Technology	Responsibility
Frontend	React / TypeScript / Tailwind	Dashboard, upload, charts and results
Backend	Node.js / Express	Auth, validation, APIs and orchestration
Document Processing	PDF/DOCX parser	Text extraction and normalization
AI Layer	LLM + prompt pipelines	Extraction, matching and recommendations
Database	PostgreSQL	Users, profiles, jobs and analyses
Storage	Object storage	Resume files and generated artifacts
Security	JWT/OAuth + encryption	Authentication and authorization

10. Major Modules

Dashboard

Shows overall scores, recommended roles, recent analyses and readiness.

Resume Analyzer

Parses the resume and evaluates structure, skills, evidence and ATS-oriented factors.

Job Analyzer

Reads a target job description and extracts requirements, responsibilities, skills and keywords.

Role Recommendation Engine

Ranks suitable roles using candidate skills, education, experience and project evidence.

Job Matching Engine

Measures how closely a resume aligns with a target job.

Skill Gap Analyzer

Identifies missing or weak capabilities and prioritizes them.

AI Resume Improvement

Suggests concise, relevant and evidence-based wording without fabricating candidate information.

11. User Workflow

- Open the dashboard.
- Upload the resume.
- Validate file type and size.
- Extract and normalize resume text.
- Build a structured candidate profile.
- Calculate resume and ATS scores.
- Generate recommended roles.
- Paste a target job description or select a role.
- Extract job requirements and keywords.
- Compare candidate evidence with job requirements.
- Display match score, gaps and recommendations.
- Review and apply appropriate changes before submitting an application.

12. Scoring Framework

The MVP can use transparent weighted scoring. A production version can combine deterministic rules with semantic AI matching. Scores are guidance indicators, not guarantees of hiring outcomes.

Metric	Prototype Example	Factors
Resume Score	82 / 100	Structure, clarity, completeness and evidence
ATS Score	88%	Keywords, sections and machine readability
Profile Score	79%	Skills, projects and qualifications
Best Role Match	91%	Highest-fit recommended role
Job Match	84%	Overall target-job alignment
Skills Match	89%	Required skills represented
Keyword Match	81%	Relevant terms represented
Experience Match	76%	Relevant responsibility evidence

13. AI Processing Pipeline

- 1. Ingestion — receive and validate the uploaded document.
- 2. Extraction — convert PDF/DOCX into normalized text.

- 3. Section detection — identify summary, skills, education, experience, projects and certifications.
- 4. Entity extraction — identify technologies, job titles, dates, organizations and qualifications.
- 5. Normalization — map equivalent skills and titles.
- 6. Job analysis — extract requirements from the target description.
- 7. Semantic matching — compare candidate evidence with job requirements.
- 8. Recommendation generation — produce targeted improvement guidance.
- 9. Validation — prevent unsupported claims.
- 10. Presentation — render results in the dashboard.

14. Example Role Recommendations

Role	Match	Example Reason
Data Analyst	91%	Python, SQL, Power BI and analytical project exposure
Python Developer	86%	Python programming and development exposure
Business Analyst	78%	Analytical and business-oriented capabilities
ML Engineer	71%	Machine learning exposure with potential experience gaps

15. Job Description Analyzer

The Job Analyzer allows the user to paste a target job description. The system extracts required and preferred skills, responsibilities, experience expectations, qualifications, tools and important keywords. These requirements are compared with the candidate profile.

Output	Prototype Example
Overall Match	84%
Skills Match	89%
Keyword Match	81%
Experience Match	76%
Missing Skills	Advanced Statistics, Tableau, ETL

16. Resume Improvement Engine

A major differentiator is the ability to explain exactly what should change for a particular role. A generic sentence such as “Worked on data analysis projects using Python” should be improved by identifying the actual dataset, method, technology, result and business impact.

The engine must never invent achievements, metrics, employers, projects, certifications or skills. Suggested content should be grounded in information already provided by the candidate and should require user confirmation.

17. User Interface and UX

- Modern SaaS-style dashboard.
- Persistent navigation: Dashboard, Resume Analysis, Job Analyzer, Recommended Roles and Skill Gap.
- Score cards for fast understanding.

- Progress bars for role and job matching.
- Skill chips for quick scanning.
- Clear visual separation of strengths, gaps and actions.
- Responsive desktop, tablet and mobile layouts.
- Consistent spacing, typography, icons and color semantics.
- Clear empty states, loading states and error messages.

18. Suggested Data Model

Entity	Key Fields
User	id, name, email, authentication data, created_at
Resume	id, user_id, file_url, version, extracted_text, created_at
CandidateProfile	user_id, skills, education, experience, projects, certifications
JobDescription	id, user_id, title, company, description, requirements
Analysis	id, resume_id, type, scores, results, created_at
RoleMatch	analysis_id, role_name, score, explanation
SkillGap	analysis_id, skill, priority, recommendation
Recommendation	analysis_id, section, original_text, suggested_text, rationale

19. Suggested REST API

Method	Endpoint	Purpose
POST	/api/resumes/upload	Upload and process resume
GET	/api/resumes/:id	Retrieve resume information
POST	/api/resumes/:id/analyze	Run resume analysis
POST	/api/jobs/analyze	Analyze a job description
POST	/api/match	Compare resume with job
GET	/api/roles/recommendations	Get recommended roles
GET	/api/skill-gaps/:analysisId	Retrieve skill gaps
POST	/api/recommendations	Generate resume suggestions

20. Security and Privacy

- Use HTTPS for all communication.
- Never expose AI provider API keys in frontend code.
- Use authentication and authorization for protected resources.
- Validate file extension, MIME type and size.
- Scan uploaded files where appropriate.
- Encrypt sensitive data in transit and at rest.
- Restrict users to their own resumes and analysis results.
- Define retention and deletion policies.

- Avoid logging unnecessary resume content.
- Provide transparent privacy and data-processing notices.

21. Testing Strategy

Area	Testing Examples
Unit	Scoring, parsing, validation and matching calculations
Integration	Upload → parse → analyze → store → display
API	Authentication, authorization, validation and response formats
UI	Navigation, upload, responsiveness and result rendering
AI Evaluation	Extraction accuracy, match relevance and recommendation quality
Security	File abuse, access control, injection and secret exposure
Performance	Large files, concurrent analysis and API latency
UAT	End-to-end candidate resume-to-application workflow

22. Error Handling

- Unsupported format: show supported file types.
- Oversized file: explain the maximum size.
- Corrupted document: request a valid copy.
- Insufficient content: identify sections that need improvement.
- AI timeout: allow retry without losing the uploaded file.
- Weak job description: request additional content.
- Low-confidence extraction: mark uncertain fields for review.
- Unauthorized request: deny access without revealing private data.

23. Deployment Strategy

- Development environment using VS Code and Git.
- Automated linting, unit tests and build checks.
- Staging environment for QA and AI evaluation.
- Separate frontend, API, database and storage services in production.
- Application monitoring for uptime, failures and analysis latency.
- Database backups and documented recovery procedures.
- Environment variables and secret-management service for sensitive configuration.

24. Future Enhancements

- AI-powered interview preparation based on the target role.
- Job discovery and personalized job recommendations.
- Application tracking dashboard.
- Multiple resume versions for different career paths.

- Evidence-based cover-letter generation.
- LinkedIn/profile optimization.
- Personalized learning roadmap for skill gaps.
- Recruiter workflows with consent and privacy controls.
- Analytics showing which resume changes improve matching.
- Multilingual resume analysis.
- Continuous profile optimization as new skills are added.

25. Expected Benefits

Benefit	Expected Outcome
Resume Quality	More targeted, readable and evidence-based resumes
Role Targeting	Better understanding of suitable positions
Time Savings	Less manual resume-to-JD comparison
Skill Awareness	Prioritized learning and improvement areas
Application Confidence	Clear actions before applying
Scalability	Consistent analysis for many users

26. Limitations and Responsible Use

- AI scores do not guarantee interviews or employment.
- Incomplete resumes can reduce matching accuracy.
- Semantic matching does not replace human judgment.
- Users should review every generated recommendation.
- The system must not fabricate candidate credentials.
- Sensitive personal information should be minimized and protected.
- The candidate retains final control over all submitted resume content.

27. Project Deliverables

- Responsive web application.
- Professional UI/UX design.
- Resume upload and parsing module.
- Resume scoring and ATS evaluation.
- Job-description analyzer.
- Role recommendation engine.
- Resume-to-job matching engine.
- Skill-gap analysis.
- AI recommendation module.
- Secure backend REST API.

- Database schema and migrations.
- Automated test suite.
- Deployment configuration.
- User and technical documentation.

28. Conclusion

The AI Resume Ranking & Job Matching System provides an end-to-end approach to resume intelligence and career optimization. It helps candidates understand their current profile, identify suitable roles, compare themselves against real job requirements and take specific actions to improve their applications.

The architecture is intentionally modular so the current website can grow into a production-grade career technology platform. The most important implementation priorities are secure document handling, transparent scoring, reliable AI evaluation, strong privacy controls and recommendations that remain faithful to the candidate's verified experience.

29. Prototype Navigation

Screen	Purpose
Dashboard	Overview of scores, recommendations and activity
Resume Analysis	Detailed resume and ATS analysis
Job Analyzer	Target job-description analysis
Recommended Roles	Ranked roles and fit percentages
Skill Gap	Strengths, missing skills and learning priorities

End of Project Documentation